ERwait: Forecasting ER Wait Times

MScAC internship project

Toronto hospital network (fictional)

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The Problem

- Emergency room waits are long and unpredictable
- ~1 in 20 patients leave without being seen
- Goal: forecast the next 4 hours of wait time, per hospital

Headline Result

Our model is 40% better.

Data

- 2 years of triage records: 1.2M visits
- Features include arrival time, triage level and time to first physician
- We removed outliers

Method

- Gradient-boosted trees
- LightGBM (released by Google in 2020)
- Rolling-window validation

Feature Engineering

- Lagged wait times (1h, 2h, 4h, 24h, 168h)
- Rolling arrival counts by triage level (CTAS 1-5)
- Statutory holidays, long weekends and school holidays
- Temperature, precipitation, snowfall and heat warnings
- Triage mix share per hour
- Staffing: scheduled physicians and nurses per shift
- Time to first physician
- Hour of day, day of week, month (cyclic encodings)
- Ambulance offload delays
- Inpatient census and admitted-no-bed counts
- Flu and COVID surveillance signals (weekly)
- Local events: concerts, sports, marathons
- Hospital site one-hot and site-level random effects
- Interaction terms: triage mix x staffing, weather x hour

Deployment

- Ready for deployment across all Ontario hospitals

Conclusion & Future Work

- 4-hour-ahead wait forecasts per hospital
- Strong headline result
- Future work: add real-time bed data